

TAMWORTH, Australia

WMO: 957620

Lat: 31.0742S

Lon: 150.8361E

Elev: 1299

StdP: 14.02

Time Zone: 10.00 (AUS)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 30.0	(c) 32.5	(d) 22.4	(e) 17.6	(f) 55.9	(g) 25.2	(h) 20.2	(i) 51.6	(j) 21.1	(k) 57.0	(l) 19.0	(m) 57.3	(n) 3.5	(o) 140	(p) 0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 25.1	(c) 97.0	(d) 66.6	(e) 93.6	(f) 66.0	(g) 90.7	(h) 65.8	(i) 72.0	(j) 84.3	(k) 70.7	(l) 83.3	(m) 69.4	(n) 81.9	(o) 11.2	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	113.1	74.4	67.4	105.6	74.0	65.8	99.7	73.8	36.7	83.7	35.4	83.4	34.3	81.8	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.9	(b) 19.1	(c) 16.9	(e) 24.7	(f) 101.4	(g) 1.9	(h) 3.1	(i) 23.3	(j) 103.6	(k) 22.1	(l) 105.4	(m) 21.1	(n) 107.2	(o) 19.6	(p) 109.4		
			DB													
			WB													
				23.8		74.6		2.2		2.7		22.2		76.6		20.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	76.3	75.2	70.8	63.5	55.8	50.3	48.3	51.1	57.1	63.1	68.7	72.8
	DBStd	10.93	5.11	5.14	4.51	4.87	5.09	4.93	4.10	5.20	5.39	5.55	6.31	5.47
	HDD50	202	0	0	0	1	11	55	82	47	5	1	0	0
	HDD65	2148	0	1	8	81	289	440	518	431	242	101	31	6
	CDD50	4824	814	705	644	405	189	66	29	82	218	406	560	706
	CDD65	1296	349	286	187	36	2	0	0	1	6	41	141	247
	CDH74	14567	3915	2929	1772	462	31	0	0	16	197	738	1692	2815
	CDH80	6111	1925	1331	598	62	1	0	0	1	37	205	704	1247
(14) Wind	WSAvg	7.7	9.1	8.7	8.2	7.4	6.7	6.5	6.4	7.2	7.5	8.1	8.7	8.6
(15) Precipitation	PrecAvg	26.6	4.4	2.3	1.7	1.6	2.0	1.4	1.8	1.7	1.8	2.4	2.6	2.7
	PrecMax	37.2	12.7	8.0	5.7	4.9	5.9	3.9	7.5	5.0	4.8	5.7	6.3	6.0
	PrecMin	14.1	0.2	0.2	0.0	0.0	0.0	0.1	0.2	0.0	0.1	0.5	0.2	0.1
	PrecStd	5.8	3.1	2.0	1.5	1.5	1.6	1.0	1.6	1.3	1.1	1.3	1.5	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.8	100.4	93.5	85.0	77.4	70.6	67.2	76.0	84.4	89.6	98.3	99.2
		MCWB	68.1	67.8	64.9	62.4	59.6	57.2	53.9	57.9	59.0	60.8	64.3	65.5
	2%	DB	97.0	95.1	89.3	81.8	73.8	67.1	64.3	71.2	79.4	85.3	92.2	94.0
		MCWB	66.4	68.4	64.8	61.9	57.8	55.1	51.7	54.9	58.3	59.9	63.6	65.1
	5%	DB	93.7	91.4	86.1	79.2	71.1	64.4	62.2	67.9	75.4	81.6	87.8	90.6
		MCWB	66.4	67.6	64.3	60.8	56.3	53.7	51.3	53.4	57.0	59.8	63.6	65.0
	10%	DB	90.3	88.0	83.2	76.5	68.4	61.8	59.6	64.4	71.6	77.9	83.4	87.2
		MCWB	66.5	66.7	63.9	59.9	55.2	52.2	50.3	51.6	56.3	58.9	62.3	64.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.9	74.3	73.9	74.8	63.2	60.8	58.4	60.1	63.7	67.5	69.9	71.9
		MCDB	88.1	87.0	75.8	74.8	71.0	67.0	61.4	71.9	74.4	79.5	84.6	85.5
	2%	WB	71.1	72.4	70.3	66.2	61.1	57.7	55.6	57.6	61.6	64.8	67.9	69.8
		MCDB	85.3	84.7	79.6	73.9	69.0	62.9	59.8	66.0	72.9	75.8	81.7	83.0
	5%	WB	70.1	70.8	68.4	63.6	59.0	55.5	53.2	55.4	59.7	62.8	66.3	68.5
		MCDB	83.8	83.4	78.3	73.0	66.1	60.9	59.3	64.5	70.1	74.1	79.0	81.5
	10%	WB	68.9	69.4	66.5	61.6	57.1	53.7	51.3	53.4	57.6	61.1	64.8	67.3
		MCDB	82.5	81.5	77.5	72.4	65.0	60.2	58.0	62.5	68.1	72.5	77.2	80.3
(35) Mean Daily Temperature Range	5% DB	MDBR	25.1	23.6	24.2	25.4	24.7	22.5	22.8	25.9	27.1	26.5	24.8	25.0
		MCDBR	30.4	28.6	28.0	28.5	28.0	25.5	26.8	30.7	32.9	33.0	32.0	30.3
		MCWBR	8.7	8.2	9.1	11.9	13.3	14.6	15.8	16.2	15.0	13.7	10.3	9.5
	5% WB	MCDBR	23.1	22.5	21.7	23.0	20.2	19.6	20.6	24.4	26.1	25.1	23.4	23.2
		MCWBR	8.0	8.2	10.4	12.7	10.6	12.4	13.6	14.1	13.7	12.5	9.5	9.3
(40) Clear-Sky Solar Irradiance	taub		0.356	0.352	0.326	0.308	0.288	0.283	0.278	0.290	0.323	0.337	0.358	0.354
	taud		2.516	2.546	2.609	2.620	2.623	2.645	2.639	2.584	2.478	2.474	2.467	2.505
	Ebn at Noon		312	307	305	295	287	281	288	297	301	308	309	314
	Edh at Noon		36	34	30	26	23	22	23	27	33	36	37	36
(44) All-Sky Solar Radiation	RadAvg		2266	2037	1720	1413	1104	877	1003	1304	1703	1980	2121	2268
	RadStd		148	166	134	92	74	74	69	89	123	172	217	175

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (16 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	-43	-234	+342	N/A

Nomenclature: See separate page